Program: **Diploma in Integrated Circuit Design and Fabrication**	
Course Code: **6378**	Course Title: VLSI CAD LAB
Semester: **6**	Credits: **1.5**
Course Category: **Program Elective**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To design and analyze the layouts of analog and digital circuits using the available VLSI CAD software tools (e.g., Magic, Electric, Siemens L-Edit IC, Cadence Virtuoso).

Course Prerequisites:

Topic	Course code	Course name	Semester
Implementation of basic electronic circuits	2041 3043	Basic Electronics Electronic Circuits	3
Implementation of combinational and sequential circuits	3044	Digital Electronics	3
Differential amplifier	4043	Linear Integrated Circuits	4
CMOS logic circuit and design, MOSFET structure, scaling and fabrication	4371	VLSI Design	4
Techniques in VLSI layout design	4373	CAD for VLSI	4

Course Outcomes:

On completion of the course, the students will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Design and verify the layouts of basic VLSI components.	9	Applying
CO2	Design and verify the layouts of combinational circuits.	12	Applying

CO3	Design and verify the layouts of sequential circuits.	9	Applying
CO4	Design and verify the layouts of analog circuits.	12	Applying
	Lab Exam	3	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3	2			
CO2	3		3	3			
CO3	3		3	3			
CO4	3		3	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Design and verify the layouts of basic VLSI components.**		
M1.01	Design the layout of n-well resistor, p-well resistor, p+ resistor and n+ resistor	3	Applying
M1.02	Design the layout of poly resistors and poly capacitors.	3	Applying
M1.03	Design and verify the layout of CMOS inverter.	3	Applying
CO2	**Design and verify the layouts of combinational circuits.**		
M2.01	Design and verify the layout of universal logic Gates: NAND & NOR.	3	Applying
M2.02	Design and verify the layout of advanced logic Gates: XOR.	3	Applying
M2.03	Design and verify the layout of CMOS Full adder.	3	Applying
M2.04	Design and verify the layout of 2:1 Multiplexer using MOS technology.	3	Applying

	Lab Exam 1	1.5	
CO3	**Design and verify the layouts of sequential circuits.**		
M3.01	Design and verify the layout of JK Flip-Flop using MOS technology	3	Applying
M3.02	Design and verify the layout of T Flip-Flop using MOS technology	3	Applying
M3.03	Design and verify the layout of Asynchronous 3-bit up counter using MOS technology.	3	Applying
CO4	**Design and verify the layouts of analog circuits.**		
M4.01	Design and verify the layout of the common source amplifier using MOS technology.	3	Applying
M4.02	Design and verify the layout of the common drain amplifier using MOS technology.	3	Applying
M4.03	Design and verify the layout of the differential amplifier using MOS technology.	3	Applying
	Open Ended Experiment	3	Applying
	Lab Exam 2	1.5	

**** - Suggested Open Ended Projects**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can-do open-ended experiments as a group of 3-5. There should be no duplication in experiments between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on open ended experiment of their choice.)

Text /Reference:

T/R	**Book Title/Author**
T1	VLSI CAD, Niranjan N. Manjunath Kotari Chiplunkar, Prentice Hall India Learning Pvt Ltd.(2011)
T2	ALGORITHMS FOR VLSI DESIGN AUTOMATION, Sabih H. Gerez, paperback edition (2006) from Wiley.
T3	Digital VLSI Chip Design with Cadence and Synopsys CAD Tools, Erik Brunvand
T4	Basic VLSI Design, Dauglas A.Pucknell, Kamran Eshraghian.

R1	Algorithm for VLSI physical design automation, Naveed sherwani,klouwer academic publishers.

Online Resources:

Sl.No	Website Link
1	http://opencircuitdesign.com/magic/tutorials/tut1.html
2	https://www.yilectronics.com/Tutorials/ElectricVLSI_Tutorials/Tutorial_1/ElectricVLSI_Tutorial1.html
3	https://www.egr.msu.edu/classes/ece410/mason/files/TutorialB.pdf